

1.10 Quiz: Arithmetic and Geometric Sequences — Markscheme

Calculator allowed. Total: 34 marks.

1. [Maximum mark: 8]

15, 23, 31, 39, ...

(a) $d = 23 - 15 = 8$ **A1**

(b) $u_n = u_1 + (n - 1)d$ **(M1)**

$u_n = 15 + 8(n - 1)$ **A1**

Accept $u_n = 8n + 7$, or any equivalent expression, for full marks. A candidate who writes $u_n = 15 + 8n$ earns M1A0.

FT: award A1 for a correct expression from the candidate's own d in (a).

(c) $u_{30} = 15 + 8(30 - 1)$ **(M1)**

$= 15 + 232 = 247$ **A1**

A common error is $15 + 8(30) = 255$: M0A0 (it is u_{31}). FT from the candidate's expression in (b).

(d) $15 + 8(n - 1) = 303$ **M1**

$8(n - 1) = 288 \Rightarrow n - 1 = 36 \Rightarrow n = 37$ **A1**

n is a positive whole number, so 303 is a term: the 37th term. **R1**

Accept $8n + 7 = 303$, $8n = 296$, $n = 37$. R1 requires the conclusion linked to n being a whole number; a bare " $n = 37$ " earns M1A1R0. Accept a table search that shows $u_{37} = 303$ for full marks.

2. [Maximum mark: 9]

$$u_5 = 38, \quad u_{14} = 11.$$

$$(a) \quad u_1 + 4d = 38 \quad \text{and} \quad u_1 + 13d = 11 \quad \text{M1}$$

$$9d = 11 - 38 = -27 \Rightarrow d = -3 \quad \text{A1}$$

$$u_1 = 38 - 4(-3) = 50 \quad \text{A1}$$

M1 for any valid route: both equations, $d = \frac{u_{14} - u_5}{14 - 5}$, or solving the system on the GDC. The two A1 marks are independent. FT: award the second A1 for u_1 correct from the candidate's own d .

$$(b) \quad 50 - 3(n - 1) < 0 \quad \text{M1}$$

$$3(n - 1) > 50 \Rightarrow n - 1 > 16.67 \Rightarrow n > 17.67, \text{ so } n = 18 \quad \text{A1}$$

$$u_{18} = 50 - 3(18 - 1) = -1 \quad \text{A1}$$

The question asks for the term, not n : the final A1 is for -1 . A final answer of " $n = 18$ " alone earns M1A1A0.

Accept an equation in place of the inequality, or a table search showing $u_{17} = 2$ and $u_{18} = -1$ (markscheme-spec §18). FT from the candidate's u_1 and d .

$$(c) \quad \text{Equal differences: } (4k - 3) - (2k + 1) = (5k + 1) - (4k - 3) \quad \text{M1}$$

$$2k - 4 = k + 4 \Rightarrow k = 8 \quad \text{A1}$$

$$\text{First three terms: } 17, 29, 41 \quad (d = 12) \quad \text{A1}$$

Accept the equivalent set-up $2(4k - 3) = (2k + 1) + (5k + 1)$ (the middle term is the mean of its neighbours). FT: award the final A1 for three terms correctly evaluated from the candidate's k .

3. [Maximum mark: 10]

(a)

- | | | | |
|-------|----------------|------------------------------|-----------|
| (i) | 5, 10, 20, 40 | geometric, $r = 2$ | A1 |
| (ii) | 7, 4, 1, -2 | arithmetic, $d = -3$ | A1 |
| (iii) | 2, 3, 5, 8 | neither | A1 |
| (iv) | 162, 54, 18, 6 | geometric, $r = \frac{1}{3}$ | A1 |

Each A1 needs the type and, for (i), (ii) and (iv), the correct d or r . Accept $r = 0.333$ in (iv). In (iii) the differences are 1, 2, 3 and the ratios 1.5, 1.67, 1.6, so neither is constant. $d = 3$ in (ii) (sign lost) earns A0.

(b) $r = \frac{20}{16} = 1.25$ **A1**

(c) $u_{12} = 16(1.25)^{12-1}$ **(M1)**

$= 16(1.25)^{11} = 186.26 \dots \approx 186$ **A1**

Accept 186.3 or 186.26. $16(1.25)^{12} = 233$ is u_{13} : M0A0. FT from the candidate's r in (b).

(d) From u_3 to u_6 is 3 steps of $\times r$: $r^3 = \frac{1215}{45} = 27$ **M1**

$r = 3$ **A1**

$u_1 = \frac{u_3}{r^2} = \frac{45}{9} = 5$ **A1**

Accept the equations $u_1 r^2 = 45$ and $u_1 r^5 = 1215$, divided to give $r^3 = 27$, for the M1. A candidate who uses $u_1 = \frac{u_3}{r} = 15$ earns M1A1A0. FT: award the final A1 for u_1 correct from the candidate's own r .

4. [Maximum mark: 7]

Job A: arithmetic, $u_1 = 38\,000$, $d = 2000$. Job B: geometric, $u_1 = 36\,000$, $r = 1.06$.

(a) Job A: $A_n = 38\,000 + 2000(n - 1)$ **A1**

Job B: $B_n = 36\,000(1.06)^{n-1}$ **A1**

Accept $A_n = 36\,000 + 2000n$. A common error is $r = 0.06$ or $r = 6$ in Job B: $A0$. $B_n = 36\,000(1.06)^n$ earns $A0$ (it gives \$38 160 in year 1).

(b) Job A: $A_3 = 38\,000 + 2000(3 - 1) = \$42\,000$ **A1**

Job B: $B_3 = 36\,000(1.06)^{3-1} = \$40\,449.60$ **A1**

Accept \$40 400 (3 s.f.) or \$40 450. FT: award each A1 for the value from the candidate's own expression in (a).

(c) Compare the two salaries year by year: **(M1)**

Year	Job A	Job B	
5	\$46 000	\$45 449.17	A1
6	\$48 000	\$48 176.12	

Year 6 **A1**

M1 for a valid comparison: a GDC table of both expressions, a graph, or solving $36\,000(1.06)^{n-1} = 38\,000 + 2000(n - 1)$ on the GDC ($n = 5.78\dots$). The middle A1 is for the two adjacent years (or the value $n = 5.78$). A single row ($B_6 > A_6$, so year 6) still earns full marks, with the feedback note that year 5 is what justifies “first” (markscheme-spec §18).

The answer is a whole year: “year 5.78” earns M1A1A0.